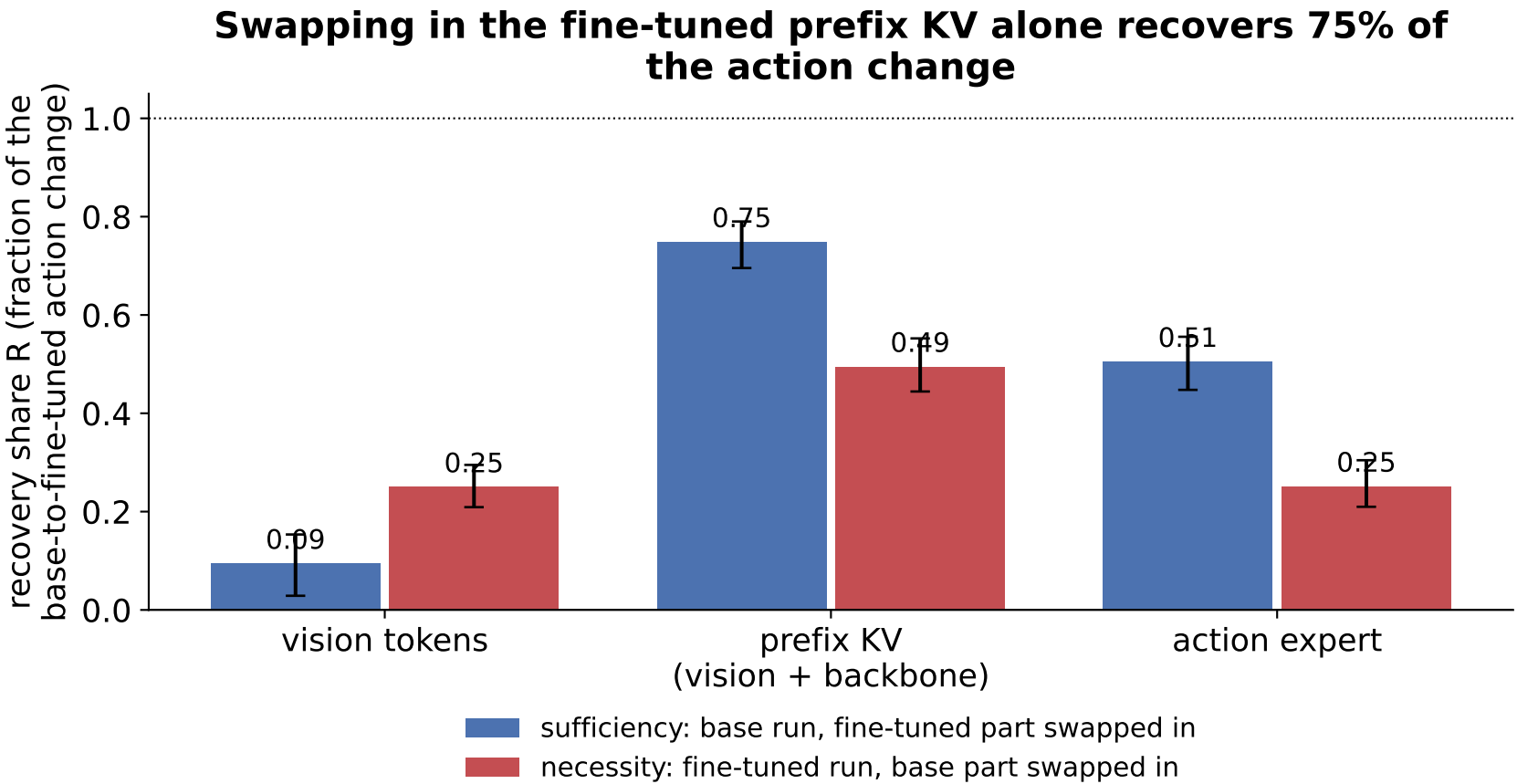

Module-level activation patching on 240 LIBERO training states; error bars are 95% bootstrap intervals over states. L is the squared error to the fine-tuned model's action chunk. By construction sufficiency(prefix KV) + necessity(action expert) = 1 and sufficiency(action expert) + necessity(prefix KV) = 1, because the expert reads only the KV cache. Pilot: pi05_base vs pi05_libero, which was fine-tuned on the full LIBERO data. This is a large fine-tune; the few-demo regime is left for later.